

Comp 442 Group Assignment 3

“Phish Detector”

Members: Bachviet Nguyen, Carlos Bautista, Finn Bishop, Haley Kenney

Dataset Overview

Dataset Info

- Email spam detection dataset
- 5,572 records
- 2 attributes
 - Category
 - Message

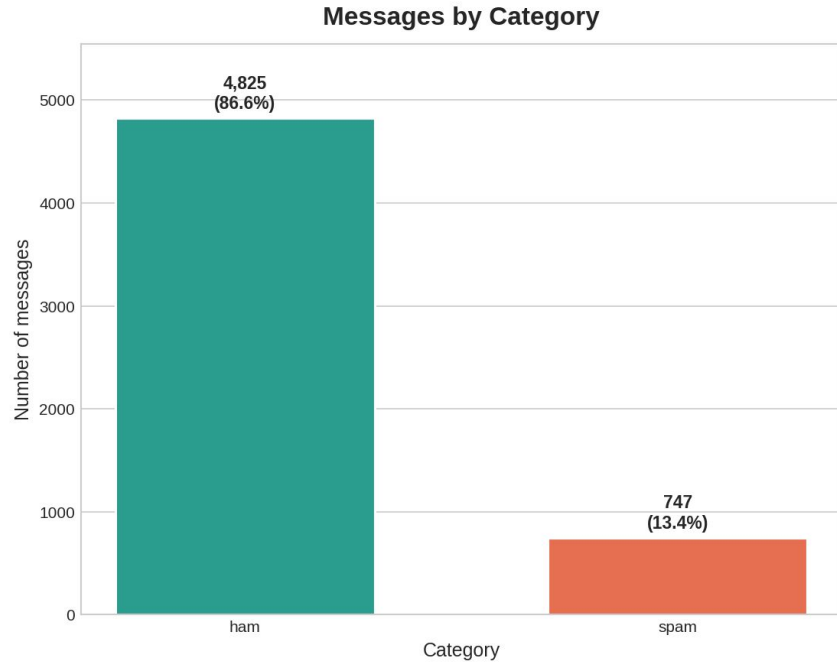
Dataset Composition

- Ham: 4,825 (86.6%)
- Spam: 747 (13.4%)

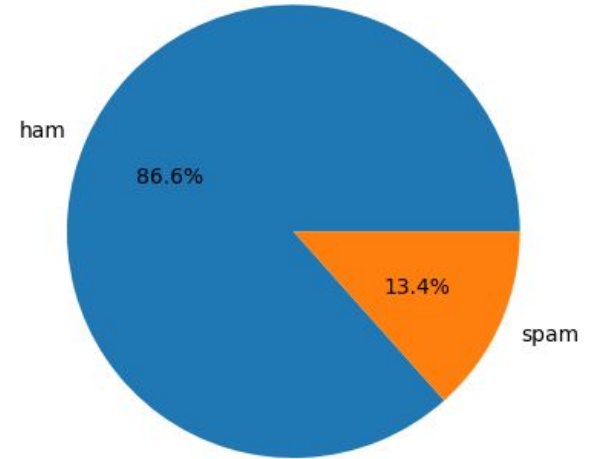
Purpose

- Explore the dataset using visualization techniques
- Identify patterns, relationships, and potential challenges for machine learning

Class Distribution



Spam vs Ham Percentage

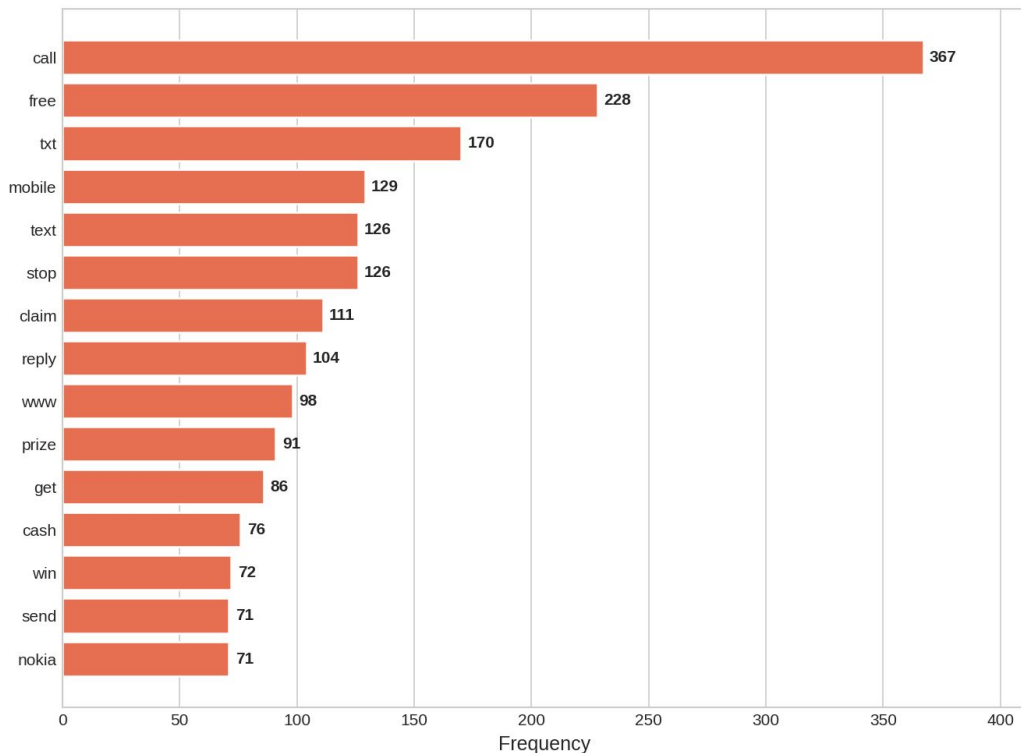


Key Findings

- Ham messages dominate the dataset
- Spam messages represent only 13.4%
- Dataset exhibits class imbalance

Common Spam Words

Most Frequent Words in Spam Messages

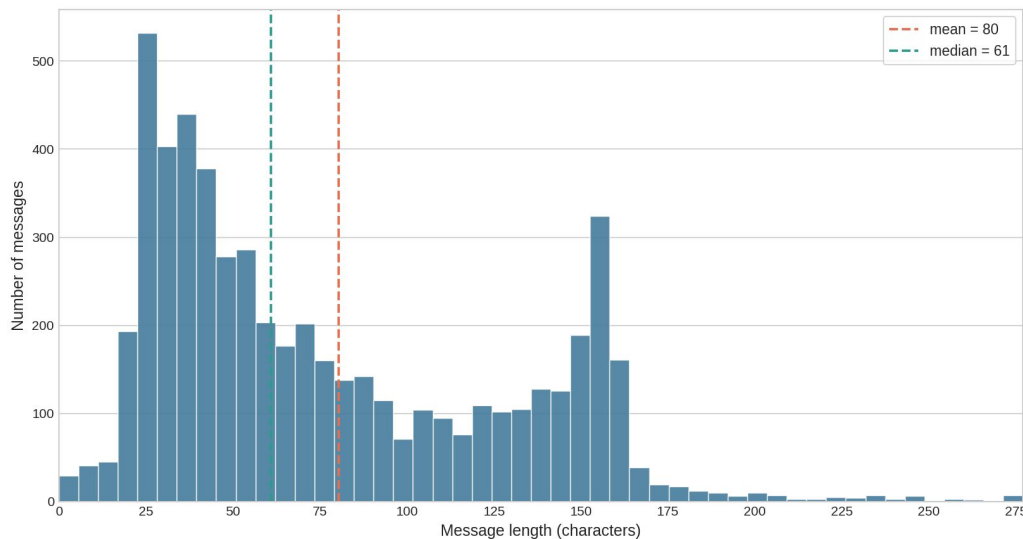


Key Findings

- Most common spam words:
 - Call
 - Free
 - Txt
 - Claim
 - Prize
- These words may serve as useful predictive features

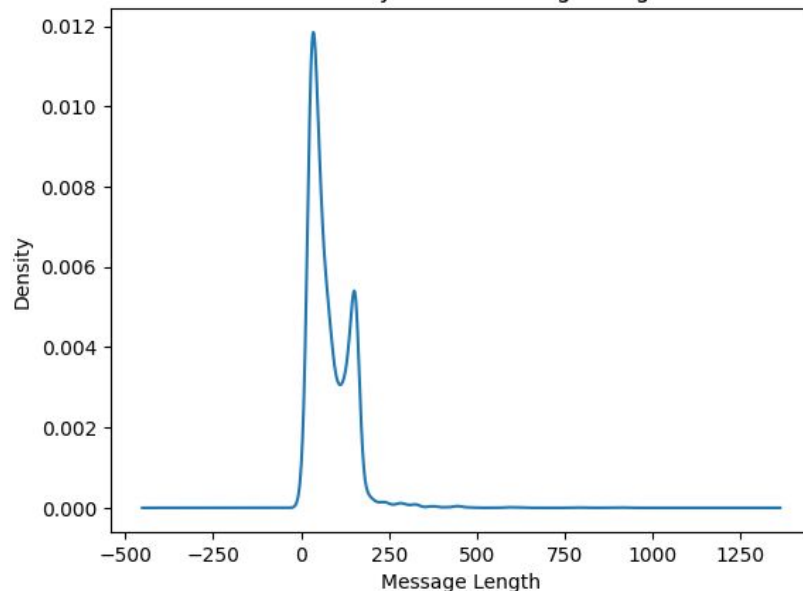
Message Length Distribution

Distribution of Message Length (all messages)



Bimodal: a tall peak of short ham messages, plus a second bump near 150 characters from spam.

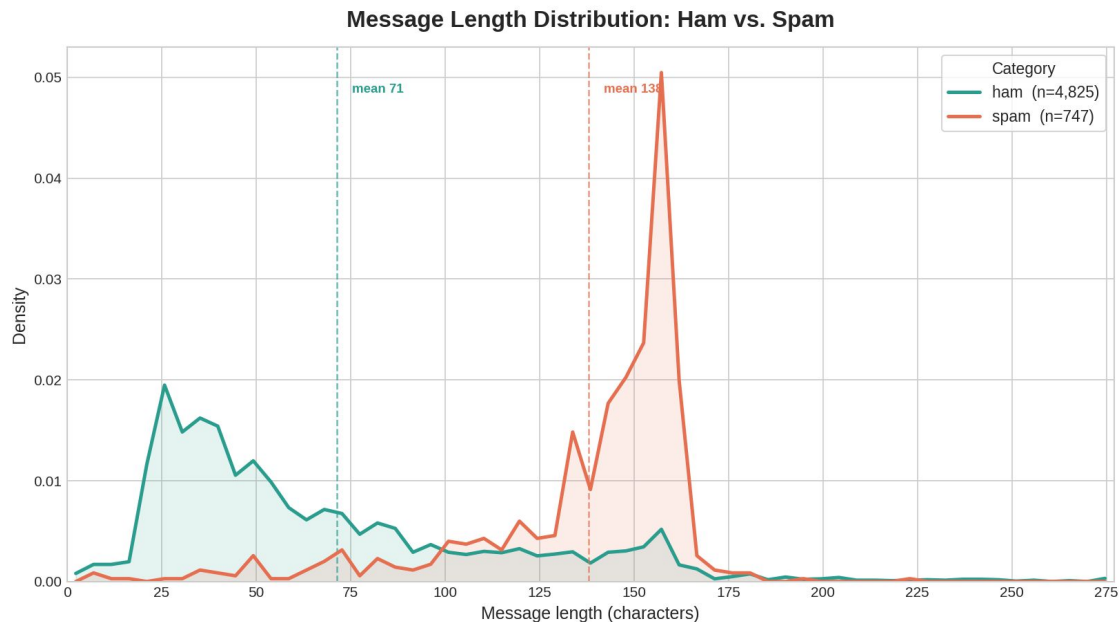
KDE Density Plot of Message Length



Key Findings

- Most messages are relatively short
- Distribution is positively skewed
- Small number of very long messages creates a right tail
- Secondary cluster around 150-160 characters

Spam vs Ham Length Comparison

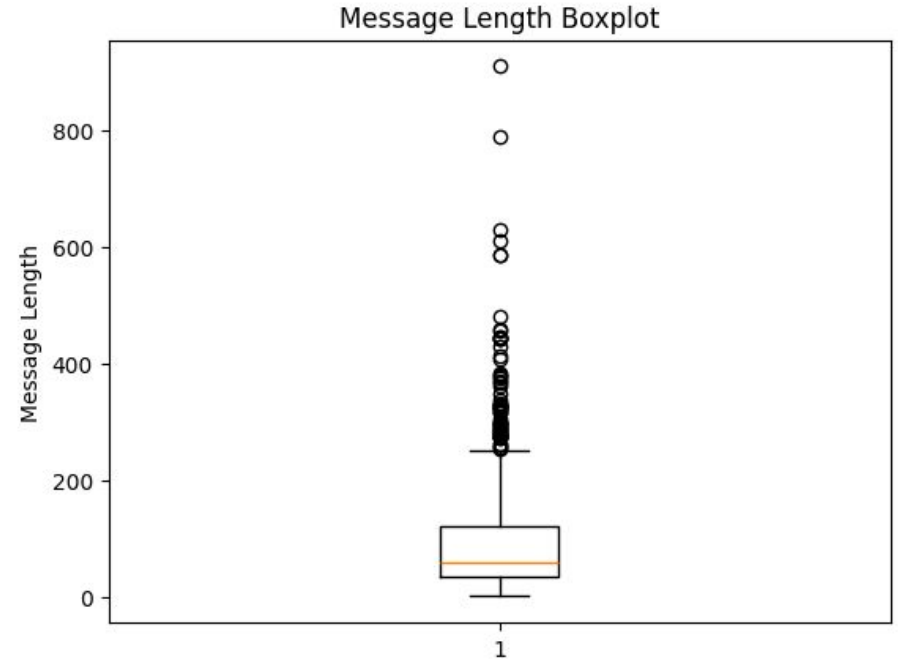
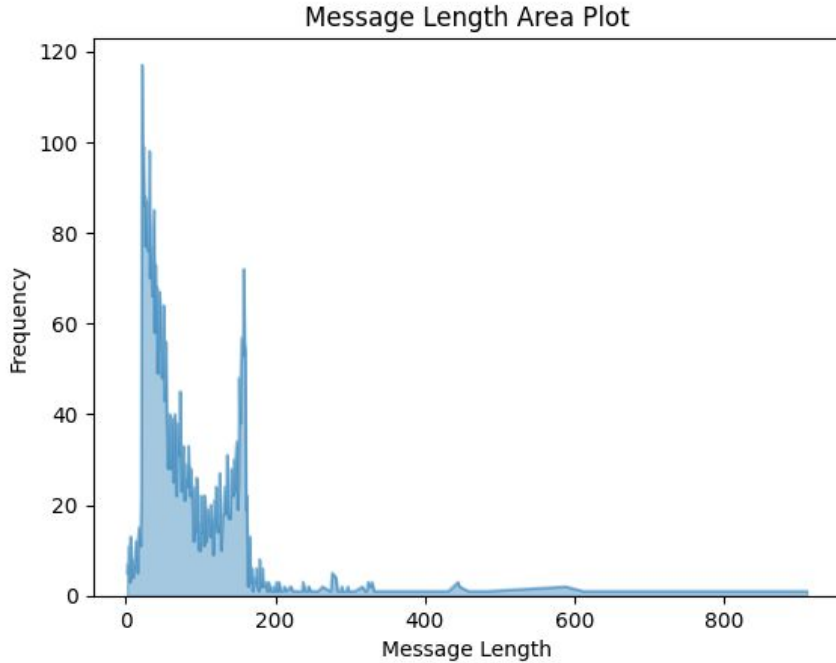


Spam clusters tightly near the 160-character SMS limit; ham is short and right-skewed.

Key Findings

- Spam messages cluster near 150-160 characters
- Ham messages are generally shorter
- Message length may help distinguish spam from legitimate messages

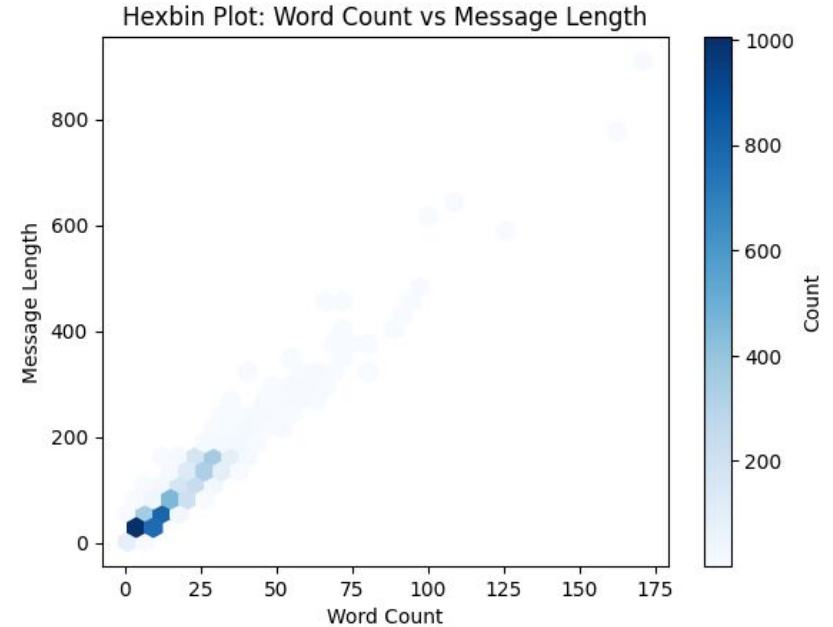
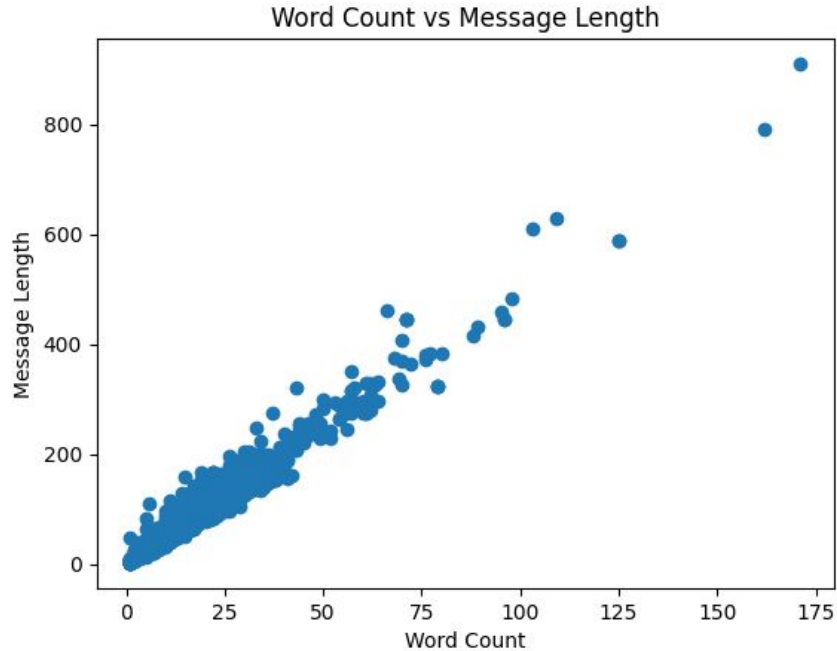
Outlier Analysis



Key Findings

- Most messages fall within a relatively small range
- Several long-message outliers are present
- Outliers contribute to positive skewness

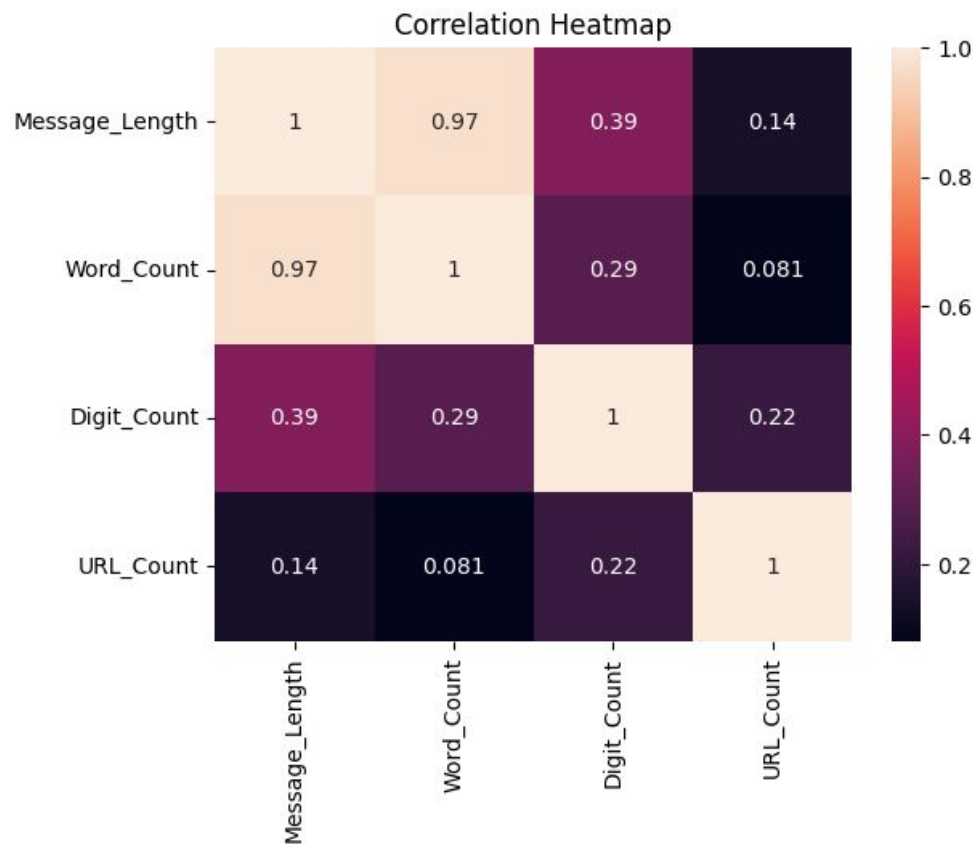
Feature Relationships



Key Findings

- Strong positive relationship between word count and message length
- Most observations cluster at lower values
- Few extreme observations exist

Correlation Analysis



Key Findings

- Message length ↔ word count = 0.97
- Digit count shows moderate relationships
- URL count has relatively weak correlations

Interpretation

- Message length and word count are highly related features

Findings and Challenges

Major Findings

- Strong class imbalance
- Positive skewness
- Presence of outliers
- Strong feature correlations

Challenges

- Duplicate records
- Class imbalance
- Region-specific language patterns
- Need for preprocessing

Conclusion

- Visualizations revealed important patterns within the dataset
- Message length and word count are highly informative features
- Spam messages show distinct characteristics compared to ham messages
- The dataset remains suitable for spam classification and future machine learning development